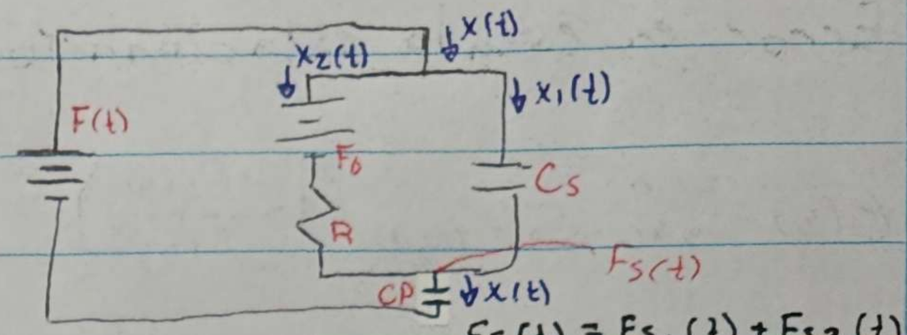
$$X(t) = x_1(t) + x_2(t)$$

- Circuito 1. Apagando $F_0(t)$



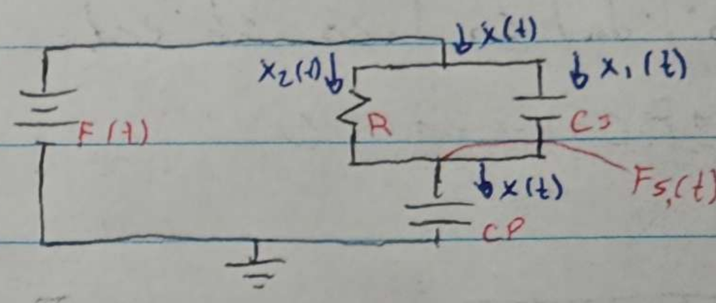
$$Fs(t) = Fs_1(t) + Fs_2(t)$$

$$X(t) = x_1(t) + X_2(t)$$

$$X(t) = C_P \frac{dF_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$



$$C_P \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_{Ps} F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(t)}{R}$$

$$C_{Ps} F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

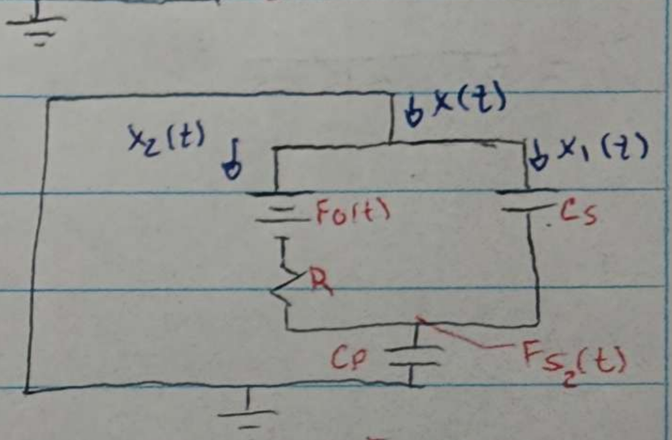
$$C_{Ps} F_{s1}(s) + C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

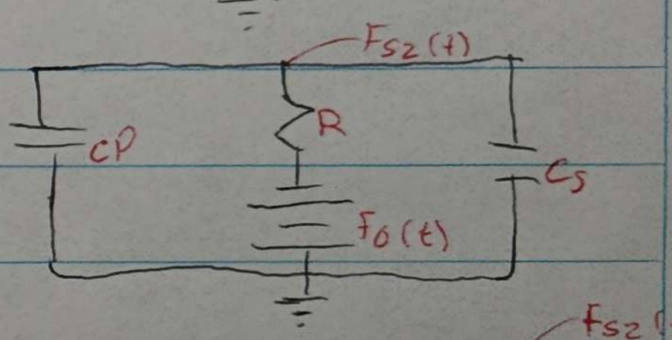
$$\left(C_{Ps} + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

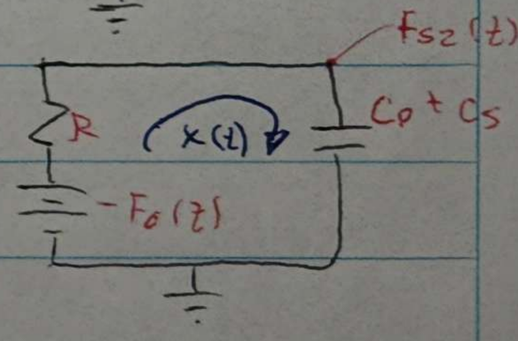
$$\frac{R(C_P + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_P + C_s)s + 1}{R}} F(s)$$

$$F_{s1}(s) = \frac{C_s R s + 1}{R(C_P + C_s)s + 1} F(s)$$







- Circuito 2. Apagando $F(t)$ $\quad F_0(t) = aF(t)$

$$-aF(t) = R x(t) + \frac{1}{C_0 + C_s} \int x(t)\,dt \Rightarrow -aF(s) = R x(s) + \frac{x(s)}{(C_P + C_s)s}$$

$$F_{s2}(t) = \frac{1}{C_P + C_s} \int x(t)\,dt \qquad F_{s2}(s) = \frac{x(s)}{(C_P + C_s)s}$$

$$F(s) = -\frac{1}{a}\left(R + \frac{1}{(C_P + C_s)s}\right) x(s) = -\frac{1}{a} \frac{R(C_P + C_s)s + 1}{(C_P + C_s)s} x(s)$$

$$\frac{F_{s2}(s)}{F(t)} = \frac{\frac{x(s)}{(C_P + C_s)s}}{-\frac{1}{a} \frac{R(C_P + C_s)s + 1}{(C_P + C_s)s} x(s)} = \frac{-a}{R(C_P + C_s)s + 1}$$

$$F_{s2}(s) = -\frac{aF(t)}{R(C_P + C_s)s + 1}$$

- Función de transferencia.

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s + 1}{R(C_P + C_s)s} F(s) - \frac{aF(s)}{R(C_P + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - a}{R(C_P + C_s)s + 1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - a}{R(C_P + C_s)s + 1}$$

- Error en estado estacionario

$$e(s) = \lim_{s \to 0} sF_{(s)} \left(1 - \frac{F_s(s)}{F(s)}\right) = \lim_{s \to 0} s \frac{1}{s} \left(1 - \frac{RCs\,S + 1 - \alpha}{(RCp + RCs)s + 1}\right) = \alpha$$

- Estabilidad en lazo abierto

$(RCp + RCs)s + 1 = 0$

$\lambda = \frac{1}{RCp + RCs}$

El sistema es Estable